

EVOLUTION

High parasite diversity accelerates host adaptation and diversification

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Host-parasite species pairs are known to coevolve, but how multiple parasites coevolve with their host is unclear. By using experimental coevolution of a host bacterium and its viral parasites, we revealed that diverse parasite communities accelerated host evolution and altered coevolutionary dynamics to enhance host resistance and decrease parasite infectivity. Increases in parasite diversity drove shifts in the mode of selection from fluctuating (Red Queen) dynamics to predominately directional (arms race) dynamics. Arms race dynamics were characterized by selective sweeps of generalist resistance mutations in the genes for the host bacterium's cell surface lipopolysaccharide (a bacteriophage receptor), which caused faster molecular evolution within host populations and greater genetic divergence among populations. These results indicate that exposure to multiple parasites influences the rate and type of host-parasite coevolution.

Coevolution between hosts and parasites is hypothesized to be central to numerous biological phenomena, from rapid evolutionary change (1) to the maintenance of diversity (2) and sexual reproduction (3, 4). Host-parasite coevolution is traditionally investigated in a pairwise framework where a single parasite species infects one host species (5). However, in nature, these pairwise relationships rarely exist in isolation (6, 7); hosts are often under attack by multiple parasite species (8), and parasites must compete for hosts (9–12). Host-parasite coevolution thus operates within a network of species interactions (13). It nevertheless remains unknown whether and how diverse parasite communities can shape coevolutionary patterns and processes.

Coevolution with highly diverse parasite communities should impose stronger selection for host resistance if this diversity yields more infections and higher mortality than single parasites. This outcome could ultimately drive faster host evolution and increased divergence among host populations. To test these predictions, we applied an experimental coevolution approach to an in vitro bacterium-phage system. We exposed the host bacterium, *Pseudomonas aeruginosa*, to communities of one to five lytic viral parasites (bacteriophages PEV2, LUZ19, LUZ7, 14-1, and LMA2) that are obligate lethal parasites. These parasites each infect their hosts by attaching to one of three specific cell surface receptors [lipopolysaccharides (LPSs), Ton-B-dependent receptors, or type IV pili], and hosts typically evolve resistance by modifying or deleting these attachment sites (14). Parasites, in turn, can evolve reciprocal adaptations to circumvent host resistance (15). We experimentally coevolved

host populations independently in cell culture tubes with replicates of a five-parasite community (high diversity, $n = 30$ cultures), with every possible pairwise parasite combination (medium diversity, $n = 30$), and in a single-parasite treatment with replicates of each parasite in monoculture (low diversity, $n = 30$), alongside a parasite-free control ($n = 30$). We tested for effects on the tempo and mode of host-parasite coevolution, conducting phenotypic assays of host resistance and parasite infectivity, as well as using deep sequencing to measure changes in the genomic compositions of host and parasite populations through time. Because we were able to experimentally manipulate parasite diversity, our work differs from the approaches used to investigate coevolution in natural communities, where diversity is not controlled exogenously (16, 17).

As with predators and prey, parasite densities are expected to track host densities, producing time-lagged cycles in population sizes over time (18). In our experiment, given that the host was initially susceptible, we might have expected large amplitude oscillations in population size, unless host resistance rapidly evolved (19). By contrast, we observed that host densities increased throughout the experiment while parasite densities remained stable (Fig. 1A and supplementary text), indicating that underlying coevolutionary dynamics interfered with the expected ecological dynamics (20).

We determined whether coevolution took place in our experiment by performing 13,500 time-shift assays. This approach assesses the infectivity of parasites toward their past, present, and future hosts and, reciprocally, the resistance of hosts to their past, present, and future parasites (21). Resistance was measured by inhibition assays in which individual host genotypes were challenged with parasite communities and resultant growth or inhibition was analyzed to compare the mean levels of resistance among treatments. Infectivity and resistance varied wide-

ly across individual host-parasite communities over time (fig. S1). Despite this complexity, two clear and significant patterns ($F_{2,13044} = 1766.21$, $P < 0.0001$) emerged. First, parasites from the future were more infective than parasites from the present or past (Fig. 1B). Likewise, hosts evolved increased resistance over time (Fig. 1C). This pattern is typical of arms race dynamics, commonly seen with in vitro bacteria-phage coevolution (5, 22). Second, bacterial resistance increased with parasite diversity and parasite infectivity decreased ($F_{2,85} = 9.7$, $P < 0.001$), suggesting that parasite communities impose stronger selection on their host than single parasites alone.

To test the hypothesis that parasite diversity accelerated coevolution at a molecular level, we sequenced entire host and parasite populations longitudinally at multiple time points (ancestor, midpoint, and end point) for the high- ($n = 12$ cultures), medium- ($n = 10$), and low-diversity ($n = 15$) and control ($n = 10$) treatments. We could thus detect any mutations that arose during coevolution and estimate their frequency changes within populations. In the hosts, we found 474 nonsynonymous and 75 synonymous polymorphisms across 173 genes and 133 intergenic mutations. Parallel evolution with mutations in the same genes or genes in the same pathways was common across all treatments: >50% of mutations occurred in <5% of these genes. Notably, we observed parallel evolution in genes that are known phage receptors (table S1), including genes for LPS ($n = 190$ mutations), type IV pili ($n = 69$ mutations), and the Ton-B-dependent receptor ($n = 55$ mutations). To measure the rate of bacterial molecular evolution, we calculated the Euclidean genetic distance of each population from the ancestor (Fig. 2A). We found that higher levels of parasite diversity drove greater host divergence from the ancestor and consequently faster host evolution (fig. S2) [analysis of variance (ANOVA), $F_{2,34} = 10.5$, $P < 0.001$; Tukey post hoc test, $P < 0.05$].

Two mechanisms could explain why diverse parasite communities accelerate the rate of host evolution. On one hand, diverse communities are more likely, by chance alone, to contain parasites that drive rapid evolution. On the other hand, it is possible that parasite diversity per se accelerates host evolution by increasing selection for resistance. To discriminate between these mechanisms, we calculated the contribution of parasite diversity to the speed of host evolution by adopting an approach developed to determine the contribution of biodiversity to ecosystem function, represented in this study by the host evolutionary rate (23). Using this approach, we found that the combination of all five parasites contributed to faster host evolution overall than phage monocultures (test for transgressive overyielding $D_{\max} > 0$; one-sample t tests with Bonferroni adjustment, $P < 0.05$). Thus, the diversity of interactions among parasites within communities, and not merely the presence of a particular parasite, imposed stronger selection.

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Whole-genome sequencing also revealed profound changes in parasite communities. At an ecological level, the composition of communities changed markedly over time (fig. S3). Parasite diversity decreased in both the medium- and high-diversity treatments, but diversity remained higher in the high-diversity treatment than in the medium-diversity treatment (fig. S4). At a genetic level, we found evidence for rapid molecular evolution of phage, which supports the results of our phenotypic assays. In total, we detected 533 single-nucleotide polymorphisms (SNPs) across all five parasite types that reached a frequency of >10% in the parasite populations (table S2). The proportions of nonsynonymous SNPs were very high, exceeding 0.7 for each parasite (table S3), and parallel evolution was common, implying that these SNPs were predominantly beneficial mutations. To investigate the role of parasite diversity in phage evolution, we focused our analysis on communities that included the parasite PEV2 because this phage was well-represented at all levels of para-

site diversity and because, compared with the genomes of the other phages used in this study, the PEV2 genome is well annotated. For example, all but two of the genes in the parasite 14-1 genome code for hypothetical proteins. We detected 225 mutations in the PEV2 genome, and the rate of PEV2 evolution was independent of parasite diversity (fig. S5). Thirty-nine percent of the PEV2 mutations ($n = 88$) occurred in a single tail fiber gene, *gp52*. The tail fibers are the means by which the parasite recognizes and binds to its receptor and are known to undergo coevolution (24). We detected mutations in *gp52* and at least one host LPS mutation (usually in *wzz* or *migA*) in every community containing PEV2 across all levels of parasite diversity (figs. S6 and S7). Collectively, these results provide genetic confirmation for reciprocal coevolution between host and parasite across all levels of parasite diversity, and they provide further evidence that parasite attachment to LPS is a key mechanism underpinning coevolution between PEV2 and *P. aeruginosa*.

Coevolutionary dynamics are usually described as being one of two types. Arms races occur when selection results in directional increases in host resistance and parasite infectivity (25, 26), and with Red Queen dynamics, negative frequency-dependent selection causes fluctuations in allele frequencies (27, 28). It has been previously shown that each parasite follows a different coevolutionary trajectory when interacting with *P. aeruginosa* in a pairwise fashion (fig. S8) (29). Using a population genomic approach, we therefore sought to examine whether parasite diversity altered the mode of coevolution.

We examined the type of coevolution occurring in each community by comparing the changes in frequency of each host allele between the mid- and end points of the experiment (29, 30). These time frames were previously shown to be suitable for observing coevolutionary dynamics in *P. aeruginosa*-phage interactions (29). We focused our analysis on the host because changes in the frequency of phage alleles in

Fig. 1. Ecological and coevolutionary dynamics of host-parasite interactions. (A) We tracked the population dynamics of hosts (solid lines) and parasites (dashed lines) for high-, medium-, and low-diversity populations alongside the parasite-free control. All plotted points show the mean population density $\pm$ SE. Bacterial populations challenged with phage rapidly recovered their density. We did not observe cyclical oscillations in host and parasite densities. CFU, colony-forming units; PFU, plaque-forming units. **(B and C)** To directly test for coevolution, we used 13,500 time-shift assays to measure changes in phage infectivity and bacterial resistance to phage. Lower levels of parasite infectivity (B) and higher levels of host resistance (C) evolved with increasing parasite diversity ($F_{2,85} = 9.7$, $P < 0.001$). Furthermore, hosts were more susceptible to parasites from the future than to contemporary parasites, and they were most resistant to parasites from the past. Likewise, future parasites were more infective than past or contemporary parasites ($F_{2,13044} = 1766.21$, $P < 0.0001$). Error bars show 1 SE.

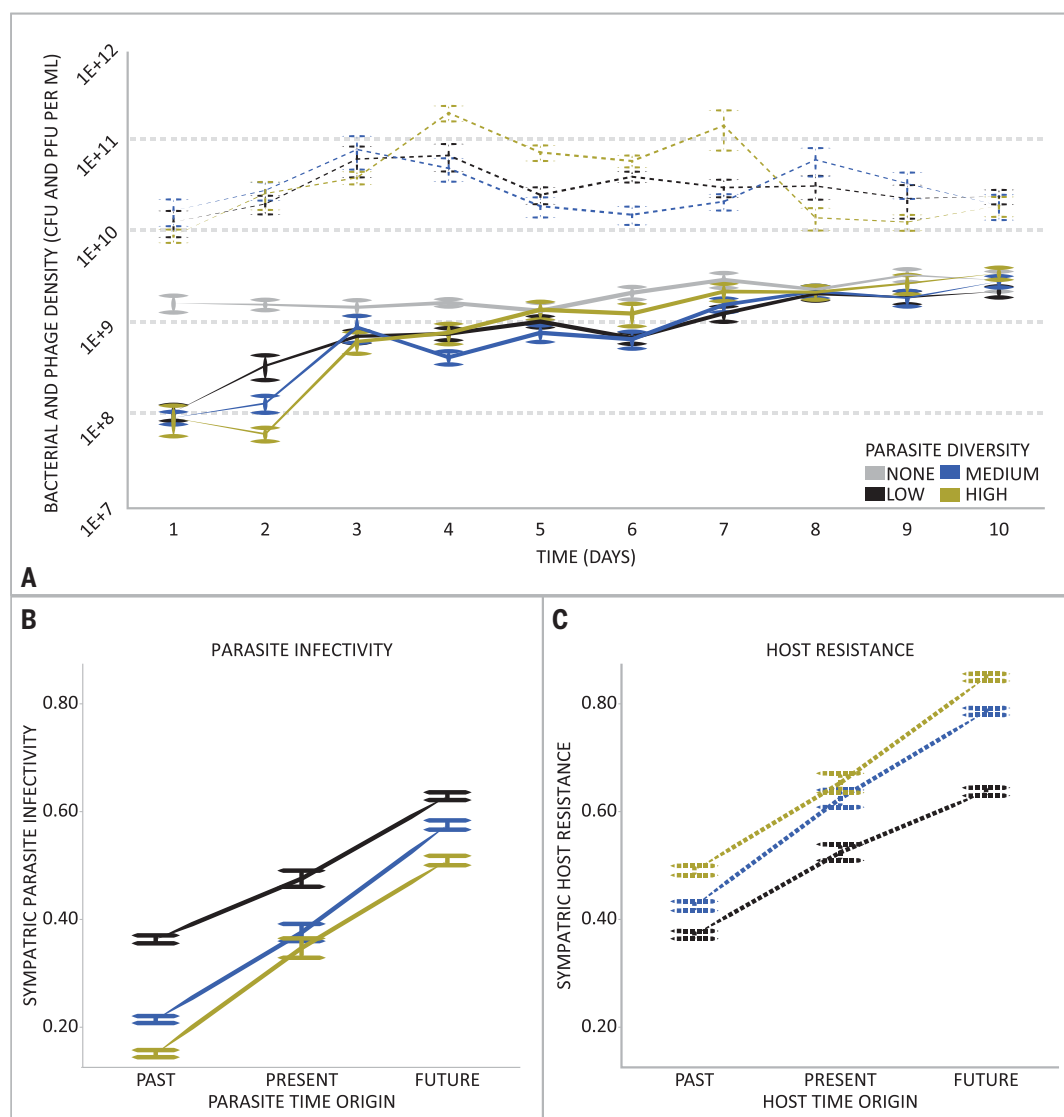


Fig. 2. Parasite diversity accelerates host evolution.

(A) Host allele frequencies after 10 days of coevolution with the high-, medium-, and low-parasite-diversity treatments were used to calculate pairwise Euclidean distances between the ancestral sequence (ANC) and each co-evolved population. Increasing parasite diversity accelerated the rate of host evolution. (B) The genetic distances between coevolved populations were ordinated by nonmetric multidimensional scaling (MDS1 and MDS2). The ellipses represent a 95% confidence bubble around the means for the different treatments. We found evidence of divergence between populations within treatments (ANOSIM, $R = 0.11$, $P < 0.01$), and the greatest within-treatment diversification was observed in the high-parasite-diversity treatment.

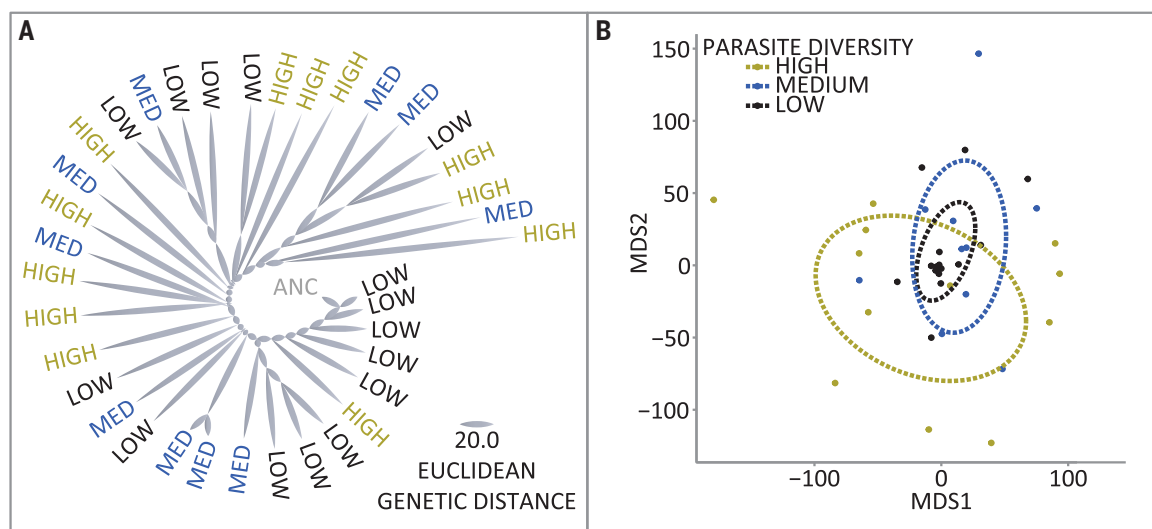
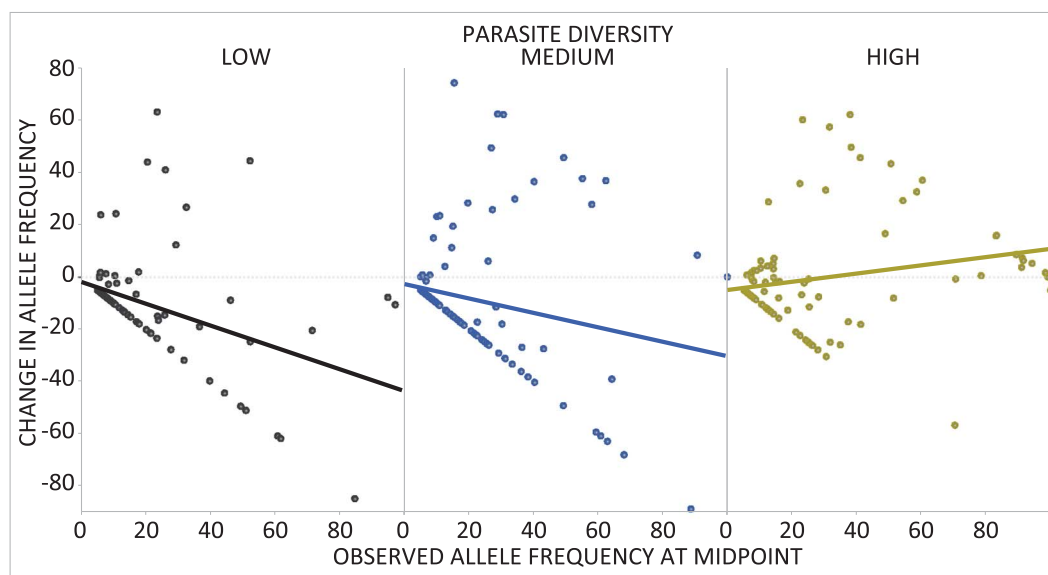


Fig. 3. Red Queen coevolution is more common in pairwise host-parasite interactions.

We tested for Red Queen dynamics by regressing the change in host allele frequency (%) from day 5 to day 10 (y axis) with the observed frequency (%) on day 5 (x axis). We found evidence for negative frequency-dependent selection on host alleles under low and medium parasite diversity, but not at high parasite diversity. The string of points forming a straight downward slope from zero in all three panels represents alleles observed on day 5 that had subsequently decreased in frequency to below our ability to detect them at day 10.



diverse communities might additionally be driven by interspecific competition among parasites (14). On average, host alleles with higher frequencies tended to decline in frequency over time in the low- and medium-parasite-diversity treatments, consistent with negative frequency-dependent selection on host resistance genes (Fig. 3). This mode of selection became less predominant at higher levels of parasite diversity [analysis of covariance (ANCOVA), $F_{2,335} = 12.64$, $P < 0.0001$], suggesting that the occurrence of Red Queen dynamics was reduced in these communities.

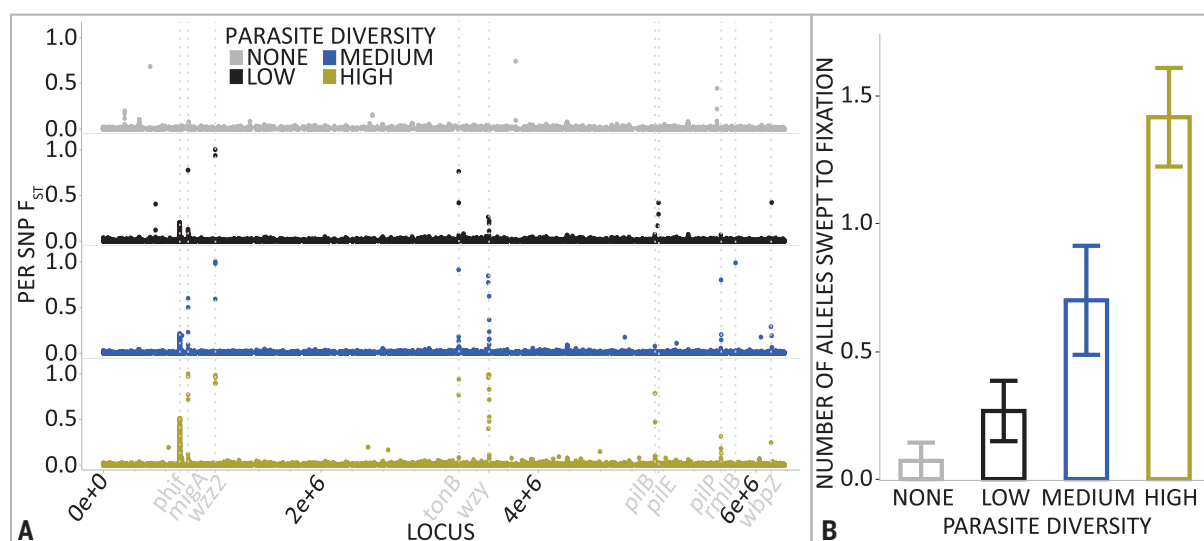
In contrast to Red Queen dynamics, arms races should result in recurrent selective sweeps of resistance mutations (5). The effects of directional selection in host-parasite interactions

have been studied in a variety of systems and can be important for coevolution of human pathogens with human populations (31–34). To test for selective sweeps in host populations, we calculated F_{ST} , which measures the genetic divergence per mutation between each coevolved host population and a sequenced ancestor (35), and we set a conservative threshold F_{ST} above which any mutation was deemed to be under strong directional selection (Fig. 4A). Strong directional selection was more common as parasite diversity increased ($\chi^2 = 20$, $df = 3$, $P < 0.001$), such that the number of fixed alleles in the high-diversity treatment was more than two times that observed under medium diversity (Fig. 4B). Of the 29 alleles approaching fixation, 23 were in genes

relating to LPS biosynthesis. LPS is very abundant on the bacterial cell surface, and phage may initially bind to LPS to anchor themselves to the cell before forming an irreversible attachment to a final receptor (36). Mutations that alter LPS biosynthesis may provide a general phage resistance mechanism. We have previously shown that mutations in the LPS biosynthesis gene *wzy* provide resistance to four of the phages used in this study (14). Red Queen and arms race dynamics fall at two ends of a continuum (21): Tight interaction specificity is required for the former, and for the latter, general host resistance mechanisms should evolve to allow future hosts to resist all previous parasite genotypes. Our data point to the availability of generalist LPS biosynthesis

Fig. 4. Parasite diversity leads to more arms race coevolutionary dynamics. (A) To test for arms race dynamics, we calculated F_{ST} for all host SNPs across the *P. aeruginosa* genome for the high-, medium-, and low-parasite-diversity treatments and the parasite-free control. Each plotted point in the panel represents a single SNP, and host

genes that are known parasite targets, including genes involved in LPS biosynthesis, type IV pilus biosynthesis, and the Ton-B-dependent receptor, are shown in italics. (B) We used a conservative F_{ST} cutoff to identify selective sweeps of SNPs per host population ($\pm$ SE), and we found that increasing parasite diversity increased the rate of fixation of SNPs ($\chi^2 = 20$, $df = 3$, $P < 0.001$).



resistance mutations as being key to the transition from Red Queen to arms race dynamics as parasite diversity increases. That both dynamics can arise from host-parasite community coevolutionary interactions is consistent with studies of bacteria-phage systems conducted in natural communities (16, 17).

Parasites have been shown to play a key role in evolutionary diversification within (37–39) and among (40) host populations. Specifically, parasite-mediated selection against common host genotypes is predicted to stably maintain genetic diversity (31, 41). Host population genetic diversity should be greatest during pairwise coevolution, where we observed that Red Queen dynamics are relatively stronger. Surprisingly, we found that levels of genetic diversity within host populations were not affected by parasite diversity (fig. S9) despite the different modes of selection acting across treatments. Parasite diversity, conversely, had a profound effect on allopatric diversification among host populations [analysis of similarity (ANOSIM), $R = 0.11$, $P < 0.01$]. The greatest divergence was found in the high-parasite-diversity treatment (Fig. 2B) (ANOVA, $F_{2,213} = 122.15$, $P < 0.001$; Tukey post hoc test, $P < 0.05$). The simplest explanation for this result is that rapid evolution within host populations leads to accelerated divergence among populations.

Our study reveals that parasite communities can form hotbeds of rapid antagonistic coevolution. Diverse parasite communities imposed stronger selection on host populations than single parasites, causing faster selective sweeps of generalist resistance mutations and higher levels of host resistance. The multispecies coevolutionary process ultimately accelerated the rate of molecular evolution within host populations and increased the genomic

divergence among populations. Host resistance to parasite infection can be specialized (26). However, broad-spectrum defense strategies that confer protection against different enemies are widespread in animals (42–44) and plants (45, 46), as well as bacteria (47–49). We should thus consider looking beyond the pairwise framework toward parasite communities as drivers of coevolution and rapid host evolution in nature. This focus could be particularly informative in biodiversity hotspots with elevated evolutionary and speciation rates (27), where a multitude of parasites might be the fuel.

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Data and materials availability: The host and parasite sequences reported in this paper have been deposited in the National Center for Biotechnology (NCBI) database (www.ncbi.nlm.nih.gov; accession numbers PRJNA448325 and PRJNA448370, respectively).

All other data are available in the supplementary materials.

SUPPLEMENTARY MATERIALS

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Tables S1 to S4
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High parasite diversity accelerates host adaptation and diversification

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Multiple parasites speed host evolution

Virtually all organisms are parasitized by multiple species, but our current understanding of host-parasite interactions is based on pairwise species interactions. Betts *et al.* address this by using the bacterium *Pseudomonas aeruginosa* and five different phage virus parasites. Increasing parasite diversity accelerated the rate of host evolution, driving both faster genomic evolution within populations and greater divergence between populations. Thus, different parasite loads prompt different evolutionary dynamics and profoundly shape host evolution by different mechanisms.

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